PASSWORD BASED GATE LOCK SYSTEM

Arduino Uno — Complete Working Code

```
/*
=========================================================
        PASSWORD BASED GATE LOCK SYSTEM
=========================================================
Hardware:
- Arduino Uno
- 4x4 Matrix Keypad
- 16x2 LCD
- Servo Motor
- Buzzer
Pin Configuration:
Keypad : D2-D8 and D11
Servo  : D9
Buzzer : D10
LCD    : A0-A5
Default Password: 1234
Press # to submit password.
Press * to clear the entered password.
=========================================================
*/
#include <Keypad.h>
#include <LiquidCrystal.h>
#include <Servo.h>
// -------------------- LCD --------------------
LiquidCrystal lcd(A0, A1, A2, A3, A4, A5);
// -------------------- SERVO --------------------
Servo gateServo;
#define SERVO_PIN 9
#define LOCK_ANGLE 0
#define OPEN_ANGLE 90
// -------------------- BUZZER --------------------
#define BUZZER_PIN 10
// -------------------- 4x4 KEYPAD --------------------
//
// 1  2  3  A
// 4  5  6  B
// 7  8  9  C
// *  0  #  D
//
const byte ROWS = 4;
const byte COLS = 4;
char keys[ROWS][COLS] =
{
  {'1', '2', '3', 'A'},
  {'4', '5', '6', 'B'},
  {'7', '8', '9', 'C'},
  {'*', '0', '#', 'D'}
};
byte rowPins[ROWS] = {2, 3, 4, 5};
byte colPins[COLS] = {6, 7, 8, 11};
Keypad keypad = Keypad(
  makeKeymap(keys),
  rowPins,
  colPins,
  ROWS,
  COLS
);
// -------------------- PASSWORD --------------------
String correctPassword = "1234";
String enteredPassword = "";
// ==============================================
// SETUP
// ==============================================
void setup()
{
  pinMode(BUZZER_PIN, OUTPUT);
  // Attach servo
  gateServo.attach(SERVO_PIN);
  // Start with gate locked
  gateServo.write(LOCK_ANGLE);
  // Initialize LCD
```

```
  lcd.begin(16, 2);
  lcd.clear();
  lcd.setCursor(0, 0);
  lcd.print("PASSWORD GATE");
  lcd.setCursor(0, 1);
  lcd.print("LOCK SYSTEM");
  delay(2000);
  showPasswordScreen();
}
// ====================================================
// MAIN LOOP
// ====================================================
void loop()
{
  char key = keypad.getKey();
  if (key == NO_KEY)
  {
    return;
  }
  // Number key pressed
  if (key >= '0' && key <= '9')
  {
    if (enteredPassword.length() < 10)
    {
      enteredPassword += key;
      lcd.setCursor(0, 1);
      lcd.print("                ");
      lcd.setCursor(0, 1);
      // Hide password using *
      for (int i = 0; i < enteredPassword.length(); i++)
      {
        lcd.print("*");
      }
    }
  }
  // # = ENTER
  else if (key == '#')
  {
    checkPassword();
  }
  // * = CLEAR
  else if (key == '*')
  {
    enteredPassword = "";
    showPasswordScreen();
  }
}
// ====================================================
// PASSWORD SCREEN
// ====================================================
void showPasswordScreen()
{
  lcd.clear();
  lcd.setCursor(0, 0);
  lcd.print("Enter Password:");
  lcd.setCursor(0, 1);
  lcd.print("                ");
}
// ====================================================
// PASSWORD CHECK
// ====================================================
void checkPassword()
{
  lcd.clear();
  // ------------------------------------------------
  // CORRECT PASSWORD
  // ------------------------------------------------
  if (enteredPassword == correctPassword)
  {
    lcd.setCursor(0, 0);
    lcd.print("ACCESS GRANTED");
    lcd.setCursor(0, 1);
    lcd.print("Gate Opening...");
    // Success beep
    tone(BUZZER_PIN, 2000, 150);
    delay(500);
```

```
    // Open gate
    gateServo.write(OPEN_ANGLE);
    // Keep gate open for 5 seconds
    delay(5000);
    // -------------------------------------------
    // LOCK GATE AGAIN
    // -------------------------------------------
    lcd.clear();
    lcd.setCursor(0, 0);
    lcd.print("Gate Closing...");
    gateServo.write(LOCK_ANGLE);
    delay(1500);
    lcd.clear();
    lcd.setCursor(0, 0);
    lcd.print("Gate Locked");
    delay(1500);
  }
  // ---------------------------------------------
  // WRONG PASSWORD
  // ---------------------------------------------
  else
  {
    lcd.setCursor(0, 0);
    lcd.print("ACCESS DENIED");
    lcd.setCursor(0, 1);
    lcd.print("Wrong Password");
    // Warning buzzer
    tone(BUZZER_PIN, 1000, 1000);
    delay(2000);
  }
  // Clear password
  enteredPassword = "";
  // Return to password screen
  showPasswordScreen();
}
/*
=========================================================
HOW TO CHANGE PASSWORD
Change this line:
String correctPassword = "1234";
Example:
String correctPassword = "5678";
=========================================================
OPERATION
1. Power ON the Arduino.
2. LCD displays the system name.
3. Enter the password using the keypad.
4. Press # to submit.
5. Correct password:
      - Access Granted
      - Servo opens the gate
      - Gate remains open for 5 seconds
      - Servo locks the gate again
6. Wrong password:
      - Access Denied
      - Buzzer sounds
7. Press * at any time while entering a password
   to clear the current entry.
=========================================================
*/
```